

Smart Recruitment Matching Platform

How to Run the Project

Step-by-step setup and execution guide based on the current repository configuration.

Backend	ASP.NET Core Web API on .NET 8
Frontend	Angular 21.2.x / TypeScript
Database	SQL Server LocalDB — database: SMRPDb
Backend URL	http://localhost:5204
Swagger	http://localhost:5204/swagger
Frontend URL	http://localhost:4200

Important: the repository configuration is the source of truth for the actual development commands and local ports. The BRD describes the intended system, while this document explains how to run the repository as currently configured.

1. Prerequisites

- Windows development machine with the required .NET and Node.js tooling installed.
- .NET 8 SDK.
- Node.js and npm. The frontend package metadata specifies npm 11.8.0 as the package manager.
- SQL Server LocalDB (MSSQLLocalDB) available on the machine.
- Visual Studio is recommended for running/debugging the ASP.NET Core backend.
- Visual Studio Code is suitable for the Angular frontend.

2. Get the Project

1. Open PowerShell or Command Prompt.
2. Clone the repository.
3. Move into the project folder.
4. Choose the branch containing the integrated implementation when following this guide.

```
git clone
https://github.com/AbiShayiny/Smart-Recruitmen
t-Matching-Platform.git cd
Smart-Recruitment-Matching-Platform
git checkout integration
```

3. Configure / Verify the Database

The backend connection string uses SQL Server LocalDB with the database name SMRPDb. You normally do not need to change it for local development unless your machine uses a different SQL Server setup.

Setting	Current repository value
Server	(localdb)\MSSQLLocalDB
Database	SMRPDb
Authentication	Trusted_Connection=True
Trust	TrustServerCertificate=True

The backend project contains Entity Framework Core tooling and migrations. Start the backend from the project directory and allow the configured database/migrations to be used. If your existing local database is already set up, keep the existing data rather than recreating it unnecessarily.

4. Run the Backend

1. Open Visual Studio and open the solution/project, or open a terminal in the Backend folder.
2. Restore .NET dependencies.
3. Build the backend to confirm the project compiles.
4. Run the API.
5. Open Swagger in the browser.

```
cd Backend dotnet restore  
dotnet build  
dotnet run
```

The configured HTTP development URL is `http://localhost:5204`. The HTTPS profile also exposes `https://localhost:7182`. Swagger launches at the `/swagger` path.

5. Run the Frontend

1. Open a second terminal and enter the Frontend folder.
2. Install the npm packages.
3. Start the Angular development server.
4. Open the frontend in a browser.

```
cd Frontend npm install  
npm start
```

The Angular development server is expected to run at `http://localhost:4200`, which is also the frontend origin allowed by the backend CORS configuration.

6. Required Login Credentials

The backend seeds a default Administrator account when no Administrator exists in the database. The repository code specifies the following administrator credentials.

Account	Email / Username	Password	Role
Default Administrator	admin@gmail.com	Admin@123	Administrator

For Job Seeker and Employer users, use the application registration flow to create accounts, then log in with the credentials you registered. The repository does not provide a separate fixed demo password for those roles in the inspected configuration.

7. Recommended Run Order

1. Start SQL Server LocalDB and make sure the database is available.
2. Start the ASP.NET Core backend.
3. Confirm that Swagger opens at <http://localhost:5204/swagger>.
4. Start the Angular frontend.
5. Open <http://localhost:4200>.
6. Register or log in as a Job Seeker or Employer for end-user flows, or use the default Administrator account for admin functions.
7. Test the main application flow: authentication → profile/company data → vacancy → matching → application → status/contact request.

8. Basic Verification Checklist

Check	Expected result
Backend build	Build completes without compilation errors.
Swagger	Swagger UI opens on the configured backend URL.
Frontend	Angular development server starts and

build/serve	page loads on localhost:4200.
Admin login	admin@gmail.com / Admin@123 can log in when the seeded Administrator exists.
Database access	Backend can connect to SMRPDb using LocalDB.
CORS	Frontend can make API requests to the backend from localhost:4200.

9. Troubleshooting Notes

- Port already in use: stop the existing backend/frontend process or use a different launch profile/port.
- Database connection error: verify SQL Server LocalDB is installed/running and that the connection string still points to (localdb)\MSSQLLocalDB.
- Swagger does not open: confirm ASP.NET Core is running with the Development environment and use the/swagger path.
- Frontend API errors: confirm the backend is running and that it is using the configured localhost:5204origin; the backend CORS policy allows localhost:4200.
- Admin login fails: verify that an Administrator record exists. The seed logic only creates the defaultAdministrator when none exists.

10. Important Security Note

The credential shown in this document is the development seed credential present in the repository. It should not be treated as a production password. Before any real deployment, change or remove seeded credentials and replace development secrets/configuration with secure environment-specific settings.

11. Source / Configuration Reference

This run guide was prepared from the repository's Backend project file, frontend package metadata, backend application settings, and launch settings. The project targets net8.0; the frontend uses Angular 21.2.x and npm 11.8.0; the database is configured as SQL Server LocalDB with database name SMRPDb; and the backend launch settings define HTTP 5204 and HTTPS 7182 with Swagger as the launch URL.